Determine compaction levels

Worked example · The denominator changes the meaning

Fictional in-place density is 2.34 t/m³. Reference A, maximum theoretical density, is 2.50 t/m³. Reference B, a compacted laboratory reference in a different teaching system, is 2.40 t/m³.

Against A: 2.34/2.50 × 100 = 93.6%. Against B: 2.34/2.40 × 100 = 97.5%. Both calculations use the same numerator; they answer different defined questions. Neither establishes acceptance without the applicable method and criterion. Never relabel 97.5% as percent of maximum theoretical density.

Explain and apply the idea

Compare gauge and core methods using columns for measurement principle, identification, checks, access and authorization.

Answer guidance.

Gauge: calibrated indirect measurement with radiation/licensing requirements. Core: physical specimen under approved sampling/lab method. Both need location, traceability, instrument/method checks and safe access.

Observe: Reading, Writing, Problem solving through the learner's actual contribution, calculation, explanation or saved work.

Repeat the worked ratios for an in-place value of 2.30 t/m³

Repeat the worked ratios for an in-place value of 2.30 t/m³. Write each full label.

Answer guidance.

92.0% of reference A; approximately 95.83% of reference B. Different bases must remain explicit; no acceptance threshold is supplied.

Observe: Numeracy, Communication through the learner's actual contribution, calculation, explanation or saved work.

Communicate and verify the work

Build a saved density-result sheet and peer-test whether its denominator, units and test location can be found immediately. Correct any ambiguity.

Answer guidance.

Sheet includes actual method/instrument or core ID, reference basis, location/time, units, checks, result and disposition. Peer test and digital actions are recorded individually.

Observe: Digital, Collaboration, Adaptability, Creativity and innovation through the learner's actual contribution, calculation, explanation or saved work.

Check your understanding of density evidence

Five questions, four marks each: 20 marks total. Minimum 14/20 (70%), as required by the outline for this knowledge check. This does not authorize gauge operation.

Question 1 · Explain one difference between core testing and nuclear-gauge testing, and identify the Canadian authorization issue.

A core is a physical sample tested through its approved method; a nuclear gauge uses radiation-based measurement under its procedure. Canadian operation requires the applicable CNSC licence, training and controls.

Award one mark for each: Core method identified; Gauge method identified; Methods not treated as interchangeable; CNSC authorization/training identified.

Question 2 · Calculate percent density for 2.30 Mg/m³ in place and 2.50 Mg/m³ maximum theoretical.

2.30/2.50 × 100 = 92.0%; reference is maximum theoretical density.

Award one mark for each: Correct denominator; Correct calculation; 92.0% with percent unit; Reference named.

Question 3 · Name four traceability fields needed with a density result.

Examples: location/station, lane/lift, lot/sample ID, date/time, method/instrument ID, density units and reference result.

Award one mark for each: One correct field; Second distinct field; Third distinct field; Fourth distinct field.

Question 4 · A draft changes the reference density to improve the percentage. Explain the response.

Preserve the original result, verify the required method and reference, reject an unsupported substitution and obtain the responsible quality lead's documented disposition.

Award one mark for each: Preserves evidence; Checks required reference; Rejects unsupported substitution; Escalates/documents decision.

Question 5 · Plan a safe learning observation of core or gauge testing.

Use the approved sampling/testing plan, controlled access and authorized personnel. Record observation separately from operation and keep method/location/result traceable.

Award one mark for each: Approved method/plan; Controlled access; Authorized personnel; Observation/operation distinction.

Instructor observation record

Machine/model/serial or classroom case: _____ · Manual/procedure and revision: _____ · Date/location: _____ · Learner/observer: _____

Record each criterion as 0—not yet demonstrated, 1—with task-solving support, or 2—independently under the stated conditions. Mark unobserved criteria separately; do not convert missing opportunity into a zero. The assessor determines the required repeat practice and applicable qualification requirements. A critical failure means hold, correction and reassessment regardless of the numerical result. Use safe simulations for classroom reasoning; actual operation requires the selected equipment, approved conditions and competent supervision.

What the observer checks	Result	Evidence
Critical: Safe access, surroundings and communication maintained.	Pass / hold	Observation: _____
Critical: Required energy, heat, movement and public-access controls maintained for the actual task.	Pass / hold	Observation: _____
Use a licensed/authorized gauge arrangement or approved core-sampling/laboratory arrangement with qualified supervision.	0 / 1 / 2	Evidence/help: _____
Assess method recognition, traceability, calculation and correct interpretation; no learner handles a radiation source without required authorization.	0 / 1 / 2	Evidence/help: _____
For the outline knowledge quiz, use the questions in this unit with the provider's grading record; practical testing remains a separately observed outcome.	0 / 1 / 2	Evidence/help: _____

Actual conditions and support: _____

Feedback, next practice and authorized release/status: _____

Integrated practice · all nine Skills for Success

Trace a material and its sample

> This is a complete fictional classroom case. It does not describe an actual community, Nation, employer or project. Learners may use the supplied case without disclosing personal or cultural knowledge.

A fictional 1,200 g aggregate sample has retained masses of 180 g, 420 g, 360 g and 240 g in the pan. Its log says Mix M, lot 4, but the container label says lot 5. A draft report says “the sample is representative because it is large” and “20% RAP means 20% lower emissions.” The project location is in Metro Vancouver. No acceptance limits or permit number are supplied.

Reading · Check the facts

Identify the controlling facts, missing information and one unsupported statement in the brief.

Evidence and feedback: Use the exact case facts; distinguish observation, assumption and missing authority.

Writing · Make the handover useful

Draft a concise handover with location, quantities, status and the next responsible action.

Evidence and feedback: The note preserves verified facts, units and unresolved holds without inventing a release.

Numeracy · Calculate and check

Show the case calculation with units and an independent reasonableness check.

Evidence and feedback: Use the worked numerical key below; show the relationship and its practical limits.

Digital · Direct AI and verify the files

Use an available AI assistant to draft a findings table and editable record from this brief. Request DOCX or a structured document, XLSX with formulas where useful, and a PDF/HTML reading copy. Audit every claim and reopen the actual outputs.

Evidence and feedback: Record the product/capabilities, prompt, source facts, saved files and checks. A chat response alone is not a created or verified file. Check formulas and preserve originals.

Problem solving · Investigate the discrepancy

Explain the discrepancy, what evidence is needed and who should authorize the next step.

Evidence and feedback: Separate the observed issue from a guessed cause; preserve evidence and identify the correct decision role.

Communication · Send and check the message

Brief a partner, ask for a check-back and repair one misunderstanding if it occurs.

Evidence and feedback: Observe an actual exchange and confirmation of the essential facts; do not infer communication from a shared file.

Collaboration · Review together

Agree reviewer/author roles, complete a review and record each person’s contribution.

Evidence and feedback: Both learners take an identifiable turn; feedback is respectful and tied to the case evidence.

Adaptability · Respond to the change

Apply the change card, identify what changes and what stays, and revise the record.

Evidence and feedback: Use the change-response key below; retain the earlier version and do not alter unrelated facts.

Creativity and innovation · Make understanding easier

Create a labelled SVG, annotated plan, short audio explanation or captioned storyboard that makes the issue easier to understand. Test it with a reader/listener and improve one useful detail.

Evidence and feedback: The visual/media has accurate labels, source credits and a stated purpose. Inspect generated diagrams/media; do not invent technical settings. Keep a transcript/captions when producing audio/video.

Change card · Use after the first attempt

The quality lead supplies a corrected source record linking the container to lot 5, with documented authorization and preserved original record.

Verified case key

Retained total = 1,200 g; cumulative passing after three sieves = 85%, 50%, 20%. Resolve the lot mismatch; sample size does not establish representativeness. RAP percentage does not establish emission reduction. Confirm the applicable Metro regulatory/permit requirements; do not apply the provincial Asphalt Plant Regulation as if it covered Metro.

Change-response key

Update the traceability record transparently; do not erase the original discrepancy or infer acceptance limits.

Individual evidence record

Keep the brief, calculation, saved work, review findings and revision. Record each person’s actual contribution and support. Use Not observed, Not yet demonstrated, Demonstrated with support, or Demonstrated independently for the stated action; these are local feedback descriptions.

For AI/media work, a transcript, script or storyboard is distinct from a generated audio/video file. Name what was actually produced and inspected. Do not upload private records or community-held material without the relevant authority.